

Bhavana K

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Professional Summary

A data-driven and results-oriented Data Engineer with around 5 years of experience in building scalable data and backend solutions using Python, SQL, and Big Data technologies. Proficient in developing robust ETL pipelines, data lakes, and warehousing solutions using tools like PySpark, Apache Airflow, Databricks, and Snowflake. Experienced in data processing, analytics, and visualization using Pandas, Tableau, Power BI, and Matplotlib. Skilled in developing RESTful APIs and microservices using Django, Flask, and FastAPI, with strong cloud experience in AWS, Azure, and GCP. Adept in working with relational and NoSQL databases including PostgreSQL, MongoDB, and Redis. Knowledgeable in CI/CD, Docker, Kubernetes, and Terraform, with a strong foundation in Agile development practices.

Skills

Programming: Python (PySpark), Scala, SQL, Hive, Spark

Cloud Platforms: Azure, AWS (S3, Glue, EMR, Redshift), Azure Synapse, Azure Data Lake Gen2, Azure Blob Storage

Data Warehousing: Snowflake, Azure SQL Database, Teradata, Cosmos DB

ETL & Orchestration: Azure Data Factory, Apache Airflow, AWS Glue, Matillion, Stream Analytics

Data Modeling: Star and Snowflake Schemas, Dimensional Modeling, Data Lake Architecture

Version Control & CI/CD: Azure DevOps, Jenkins, Git

Visualization: Power BI, Tableau

Other Tools: DBT, Kafka, REST API, Linux, Shell Scripting

Experience

UnitedHealth Group - Data Engineer | Nov 2023– Present

- Designed, developed, and maintained scalable data pipelines and ETL workflows using AWS Glue, Databricks, and Azure Data Factory, supporting the integration of EHR/EMR, claims, and provider datasets.
- Modernized legacy data systems by migrating on-prem SQL Server and Teradata platforms to AWS S3, Snowflake, and Azure Synapse, improving performance and reducing operational cost by 25%.
- Implemented Terraform for infrastructure provisioning and AWS Step Functions for orchestration, ensuring a robust, automated, and reproducible data environment.
- Built containerized microservices using Amazon ECS and Kubernetes (EKS) to streamline data ingestion and transformation across hybrid cloud environments.
- Engineered PySpark-based data transformation logic to process high-volume healthcare data (10M+ records daily), enhancing data accuracy and timeliness for analytics teams.
- Integrated Kafka streams for near real-time ingestion of patient and operational data into the Databricks lakehouse environment.
- Deployed CI/CD pipelines via Azure DevOps and GitHub Actions for automated testing, validation, and deployment of ETL workflows.
- Established proactive data quality monitoring using SQL- and DBT-based checks, reducing anomalies by 30% and improving reliability across data layers.
- Partnered with BI and analytics teams to deliver Power BI dashboards supporting claims optimization and patient outcome analysis.

Alt. Investments – Python Developer/Data Analyst | Oct 2019 – Jul 2022

- Automated data processing, file management, and system monitoring tasks using Python scripts, improving operational efficiency, and reducing manual errors.
- Optimized database operations with Sybase and MongoDB, improving query execution times and data retrieval performance.
- Designed data pipelines for structured and unstructured data processing, integrating with Elasticsearch, Big Data ecosystems, and MongoDB.
- Developed asynchronous Python scripts to fetch and process API data, store results in Excel and dashboards, and automate workflows.
- Implemented CI/CD pipelines using Jenkins, Docker, and Kubernetes, ensuring seamless deployment, scalability, and high availability.
- Assisted in migrating AutoSys jobs to Promon, restructuring job workflows and optimizing performance for Big Data processing.
- Enhanced log monitoring and real-time analytics by integrating Elasticsearch and Kibana, reducing debugging and troubleshooting time.
- Led efforts to remediate 85%+ of security vulnerabilities in legacy systems by applying secure coding practices and risk mitigation strategies.
- Collaborated with cross-functional teams to document migration strategies, best practices, and post-migration performance benchmarks.
- Strong experience in infrastructure as code (IaC) using Terraform, Ansible, and CloudFormation, ensuring automated

- provisioning and configuration management.
- Set up S3-based data lakes and AWS Glue jobs, enhancing data storage and transformation efficiency.
- Proficient in log aggregation, monitoring, and alerting using ELK stack (Elasticsearch, Logstash, Kibana), Prometheus, and Grafana, ensuring proactive issue resolution.

Education

Wilmington University (3.9/4) Master's in Computer and Information Systems	Aug 2022 – Apr 2024
Teegala Krishna Reddy Engineering College (3/4) Bachelor's in Information Technology	Jul 2017 – May 2021

Certifications

- Snowflake Core Certification
- Azure Data Engineer Associate
- Data Modeling and Data Warehouse Fundamentals